

LEI PENG

Ph. D candidate
Electrical and Computer Engineering
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EDUCATION

Ph. D candidate, Electrical and Computer Engineering

Michigan State University, East Lansing, MI, USA

SEP 2021 – PRESENT

Master of Engineering, Electrical and Electronic Engineering

ShanghaiTech University, Shanghai, China

University of the Chinese Academy of Sciences, China

SEP 2018 – JUN 2021

Bachelor of Engineering, Control Technology and Instrumentation

Hefei University of Technology, Hefei, China

SEP 2012 – JUN 2016

EMPLOYMENT

Research Assistant

Michigan State University | East Lansing, MI, USA

SEP 2021 – PRESENT

Research Intern/Engineer

ACME Industrial, Inc. | Staten Island, NY, USA

DEC 2023 – FEB 2024

Teaching Assistant

ShanghaiTech University | Shanghai, China

SEP 2019 – JUN 2020

Research Assistant

ShanghaiTech University | Shanghai, China

OCT 2017 – AUG 2018

Electrical Engineer

AAC Technologies Holdings Inc | Changzhou, China

MAY 2017 – SEP 2017

Assistant Electrical Engineer

Hefei Metalforming Intelligent Manufacturing Co., Ltd | Hefei, China

JUN 2016 – MAY 2017

RESEARCH PUBLICATIONS

([Google scholar citations: 511, h-index:12](#))

JOURNAL PAPER

- [1] L. Peng, N. Zhang, X. Huang, Z. Li, and Y. Deng, “Geometry complex structure inspection using stretchable eddy current array sensors,” *NDT E Int.*, vol. 156, p. 103482, Dec. 2025, doi: 10.1016/j.ndteint.2025.103482.
 - [2] L. Peng *et al.*, “In-situ 3D temperature field modeling and characterization using eddy current for metal additive manufacturing process monitoring,” *Sci. Rep.*, vol. 15, no. 1, p. 9999, Mar. 2025, doi: 10.1038/s41598-025-94553-6.
 - [3] L. Peng, X. Huang, G. Piao, and Y. Deng, “A novel differential excitation capacitive sensing for hydrogen pipeline inspection,” *NDT E Int.*, vol. 144, p. 103084, Jun. 2024, doi: 10.1016/j.ndteint.2024.103084.
 - [4] L. Peng, N. Zhang, Y. Tao, and C. Ye, “Fastener Tilt Assessment Based on Magnetic Field Image Obtained With Array Tunnel Magnetoresistance Sensors,” *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021, doi: 10.1109/TIM.2020.3011782.
 - [5] L. Peng, X. Cai, N. Zhang, Z. Li, and Y. Deng, “In-line Inspection for Ferromagnetic Bent Pipe Using 3D-printed Flexible MFL Sensor Array,” *IEEE Trans. Instrum. Meas.*, 2026, doi: Accepted July 2026.
 - [6] X. Huang *et al.*, “A novel multi-fidelity Gaussian process regression approach for defect characterization in motion-induced eddy current testing,” *NDT E Int.*, vol. 150, p. 103274, Mar. 2025, doi: 10.1016/j.ndteint.2024.103274.
 - [7] Z. Miles *et al.*, “Hybrid multi-modal NDE sensing system for in-motion detection and localization of rolling contact fatigue damage in rails”.
 - [8] Y. Mei *et al.*, “Simultaneous Shape Reconstruction and Force Estimation of Soft Bending Actuators Using Distributed Inductive Curvature Sensors,” *IEEE/ASME Trans. Mechatron.*, vol. 29, no. 4, pp. 2849–2857, Aug. 2024, doi: 10.1109/TMECH.2024.3397825.
 - [9] S. Mukherjee, L. Peng, L. Udpa, and Y. Deng, “Dynamic Defect Detection in Fast, Robust NDE Methods by Transfer Learning Based Optimally Binned Hypothesis Tests,” *Res. Nondestruct. Eval.*, vol. 35, no. 2, pp. 70–101, Mar. 2024, doi: 10.1080/09349847.2024.2316916.
 - [10] X. Huang *et al.*, “Fast VGG: An Advanced Pre-Trained Deep Learning Framework for Multi-Layered Composite NDE via Multifrequency Near-Field Microwave Imaging,” *Res. Nondestruct. Eval.*, vol. 35, no. 2, pp. 102–118, Mar. 2024, doi: 10.1080/09349847.2024.2319765.
 - [11] P. Probst *et al.*, “Miniaturized multi-modality field-ready sensing system for defect detection of CFRP materials,” *NDT E Int.*, vol. 137, p. 102815, Jul. 2023, doi: 10.1016/j.ndteint.2023.102815.
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- [12] N. Zhang, L. Peng, Y. He, and C. Ye, "Flexible Probe With Array Tunneling Magnetoresistance Sensors for Curved Structure Inspection," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–9, 2022, doi: 10.1109/TIM.2022.3149102.
- [13] N. Zhang, L. Peng, X. Tao, and C. Ye, "Flexible ECT probe with front-end differential setting for inspection of curved CFRP structures," *Compos. Part B Eng.*, vol. 227, p. 109404, Dec. 2021, doi: 10.1016/j.compositesb.2021.109404.
- [14] X. Tao, L. Peng, Y. Tao, and C. Ye, "Inspection of Defects in Weld Using Differential Array ECT Probe and Deep Learning Algorithm," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–9, 2021, doi: 10.1109/TIM.2021.3099566.
- [15] C. Ye, N. Zhang, L. Peng, and Y. Tao, "Flexible Array Probe With In-Plane Differential Multichannels for Inspection of Microdefects on Curved Surface," *IEEE Trans. Ind. Electron.*, vol. 69, no. 1, pp. 900–910, Jan. 2022, doi: 10.1109/TIE.2021.3050376.
- [16] Y. Bai *et al.*, "Novel array ECT probe with three-phase excitation for steam generator tube inspection," *Int. J. Appl. Electromagn. Mech.*, vol. 64, no. 1–4, pp. 789–797, Dec. 2020, doi: 10.3233/JAE-209391.
- [17] M. Wang, X. Tao, L. Peng, and C. Ye, "Imaging of Magnetic Nanoparticles Using Small TMR Sensor With an Excitation-Compensation Scheme," *IEEE Sens. J.*, vol. 20, no. 19, pp. 11082–11089, Oct. 2020, doi: 10.1109/JSEN.2020.2996741.
- [18] N. Zhang, C. Ye, L. Peng, and Y. Tao, "Eddy Current Probe With Three-Phase Excitation and Integrated Array Tunnel Magnetoresistance Sensors," *IEEE Trans. Ind. Electron.*, vol. 68, no. 6, pp. 5325–5336, Jun. 2021, doi: 10.1109/TIE.2020.2989704.
- [19] Y. Tao, L. Peng, X. Li, and C. Ye, "Eddy Current Probe With Integrated Tunnel Magnetoresistance Array Sensors for Tube Inspection," *IEEE Trans. Magn.*, vol. 56, no. 5, pp. 1–8, May 2020, doi: 10.1109/TMAG.2020.2981435.
- [20] M. Wang, Y. Wang, L. Peng, and C. Ye, "Measurement of Triaxial Magnetocardiography Using High Sensitivity Tunnel Magnetoresistance Sensor," *IEEE Sens. J.*, vol. 19, no. 21, pp. 9610–9615, Nov. 2019, doi: 10.1109/JSEN.2019.2927086.
- [21] E. R. Rosenberg *et al.*, "Multifrequency eddy-current detection of fast transient thermal signatures for *in situ* monitoring applications," *J. Appl. Phys.*, vol. 138, no. 15, p. 154501, Oct. 2025, doi: 10.1063/5.0296071.
- [22] N. Zhang, C. Ye, L. Peng, and Y. Tao, "Novel Array Eddy Current Sensor With Three-Phase Excitation," *IEEE Sens. J.*, vol. 19, no. 18, pp. 7896–7905, Sep. 2019, doi: 10.1109/JSEN.2019.2919661.
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CONFERENCE PROCEEDINGS

- [23] L. Peng, B. Edward, M. Saptarshi, T. Joe, S. David, and D. Yiming, "In-situ Temperature Monitoring Using an Eddy Current Sensor Array with Full Matrix Capture Method," in *ASNT Research Symposium 2024 Proceedings*, Dec. 2024. doi: <https://doi.org/10.32548/RS.2024.025>.
 - [24] L. Peng, M. Subrata, and D. Yiming, "Hydrogen Induced Cracking Inline Inspection using Flexible and Differential Capacitive Sensors Array," in *ASNT Research Symposium 2023 Proceedings*, Jun. 2023. doi: <https://doi.org/10.32548/RS.2024.025>.
 - [25] L. Peng, B. Edward, M. Saptarshi, M. Rosa E., T. Joe, S. David, and D. Yiming, "In-situ 3D Temperature Field Modeling and Characterization using Eddy Current for Metal Additive Manufacturing Process Monitoring," in *ASNT Research Symposium 2023 Proceedings*, Jun. 2023. doi: DOI: 10.32548/RS.2023.039.
 - [26] L. Peng, C. Ye, Y. Tao, C. Long, and M. Li, "Inspection of the undulation structure with an in-plane differential flexible array probe," in *2021 IEEE 8th International Workshop on Metrology for AeroSpace (MetroAeroSpace)*, Naples, Italy: IEEE, Jun. 2021, pp. 547–551. doi: 10.1109/MetroAeroSpace51421.2021.9511698.
 - [27] N. Zhang, C. Ye, X. Tao, L. Peng, Y. Tao, and M. Li, "Flexible PCB with Differential Array Coils for Irregular Shape Inspection," in *2021 IEEE 30th International Symposium on Industrial Electronics (ISIE)*, Kyoto, Japan: IEEE, Jun. 2021, pp. 1–6. doi: 10.1109/ISIE45552.2021.9576179.
 - [28] S. Mukherjee, E. Benavidez, J. W. Tringe, L. Peng, Y. Deng, and D. Stobbe, "An In-Situ Eddy Current Diagnostic for Temperature Monitoring in Metal Additive Manufacturing," in *2023 IEEE International Symposium on Antennas and Propagation and USNC-URSI Radio Science Meeting (USNC-URSI)*, Portland, OR, USA: IEEE, Jul. 2023, pp. 341–342. doi: 10.1109/USNC-URSI52151.2023.10237400.
 - [29] Y. D. Kotriwar, O. Elshafiey, L. Peng, Z. Li, V. Srinivasan, E. Davis, and Y. Deng, "Gradient feature-based method for Defect Detection of Carbon Fiber Reinforced Polymer Materials," in *2023 IEEE International Conference on Prognostics and Health Management (ICPHM)*, Montreal, QC, Canada: IEEE, Jun. 2023, pp. 246–252. doi: 10.1109/ICPHM57936.2023.10194165.
 - [30] M. Wang, L. Peng, and C. Ye, "Magnetocardiography Measurements with a High Sensitivity Tunnel Magnetoresistance Sensor," in *2019 IEEE International Conference on Sensors and Nanotechnology*, Penang, Malaysia: IEEE, Jul. 2019, pp. 1–4. doi: 10.1109/SENSORSNANO44414.2019.8940068.
 - [31] N. Zhang, Y. Tao, L. Peng, and C. Ye, "Array Electromagnetic Probe with Three-Phase Excitation for CFRP Inspection," in *2019 International Conference on Microwave and Millimeter Wave Technology (ICMMT)*, Guangzhou, China: IEEE, May 2019, pp. 1–3. doi: 10.1109/ICMMT45702.2019.8992179.
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PRESENTATIONS

ORAL PRESENTATIONS

- Omnidirectional eddy current testing technique with phase-amplitude controlled array. *The 28th International Workshop on Electromagnetic Nondestructive Evaluation (ENDE2025). Knoxville, Tennessee.*
- Field testing of multi-modal UT and EM for in-motion rail inspection. *ASNT 2025 Research Symposium, Indianapolis, Indiana.*
- In-line Inspection for Steel Bent Pipe Using Flexible MFL Probe with Magnetic Sensor Array. *ASNT 2025 Research Symposium, Indianapolis, Indiana.*
- In-depth Temperature Monitoring Using Eddy Current Sensor Array. *ASNT 2025 Research Symposium. Indianapolis, Indiana.*
- Stretchable Eddy current Sensors Array Probe for Geometry Complex Surface Inspection. *ASNT 2024 Research Symposium. Pittsburgh, Pennsylvania.*
- In-situ Temperature Monitoring Using an Eddy Current Sensor Array with Full Matrix Capture Method. *ASNT 2024 Research Symposium. Pittsburgh, Pennsylvania.*
- In-situ 3D Temperature Field Modeling and Characterization using Eddy Current for Metal Additive Manufacturing Process Monitoring. *ASNT 2023 Research Symposium, Columbus, Ohio.*

POSTER

- Hydrogen Induced Cracking Inline Inspection using Flexible and Differential Capacitive Sensors Array. *ASNT 2023 Research Symposium, Columbus, Ohio.*
- A Novel Capacitive Sensing Method with Differential Excitations for Hydrogen Pipeline Inspection. *DOT PHMSA's 2023 R&D Forum, Washington, DC.*
- A Novel Capacitive Sensing Method with Differential Excitations for Hydrogen Pipeline Inspection. *The Fall 2022 Meeting of the PRCI Technical Committees, Phoenix, Arizona.*

INVENTIONS

- **(US Patent pending)** System and method for eddy current manipulation with a sensor array.
 - **(US Patent pending)** Flexible inductive sensor and method for shape reconstruction and force determination.
 - **(Submitted to MSU Technologies-TEC2025-0153)** Stretchable eddy current array probe design for complex geometry non-destructive evaluation.
 - **(Submitted to MSU Technologies-TEC2025-0151)** A novel flexible sensors array design for hydrogen induced cracking inline inspection.
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AWARDS

- **FITCH H. BEACH AWARD [1st place in department of ECE]**
2025-26 FITCH H. BEACH AWARD for Outstanding Graduate Research by an Engineering PhD Student.
- **ASNT Foundation Student Travel Grant Award**
ASNT 2026 Research Symposium. Salt Lake City, Utah.
- **ASNT Foundation Student Travel Grant Award**
ASNT 2025 Research Symposium. Indianapolis, Indiana.
- **ASNT Foundation Student Travel Grant Award**
ASNT 2024 Research Symposium. Pittsburgh, Pennsylvania.
- **Top Papers Nomination**
ASNT 2025 Research Symposium. Indianapolis, Indiana.
- **2025 ASME DSCD Best Joint Journal/Conference Paper Award on Mechatronics**

RESEARCH PROJECT PARTICIPATION

- **(Key contributor)** Advanced Electromagnetic Array for In-situ AM Process Monitoring, Department of Energy, 2025-present
 - **(Key contributor)** Towards Safe and Efficient Gas Transport Pipeline Network with Learning Enabled Autonomous Risk Assessment Systems (LARAS), National Science Foundation, 2025-present. NSF grant # 2450946.
 - **(Key contributor)** Advanced in-situ processes monitoring diagnostics and ex-situ material characterization for laser-based metal AM processes, Department of Energy, 2022-2025. DOE Lawrence Livermore National Laboratory under Contract DE-AC52-07NA27344 and supported by the LLNL-LDRD Program under Project No.23-ERD-009.
 - **(Key contributor)** Knowledge guided automation for integrity management of aging pipelines for hydrogen transport, department of transportation, 2021-2024. DOE PHMSA grant number: 693JK32150004CAAP.
 - Integrated multi-wave hybrid electromagnetic and ultrasonic sensing for real-time in-motion rails fatigue and fractures detection, Association of American Railroads, MXV Rail, 2019-2025.
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ACADEMIC SERVICE

JOURNAL REVIEWER

IEEE Transactions on Industrial Informatics

IEEE Sensors Journal

IEEE Transactions on Magnetics

Results In Engineering

NDT&E International

Measurement

Process Safety and Environmental Protection

Nondestructive Testing and Evaluation

Research In Nondestructive Testing and Evaluation

Materials Evaluation

IEEE Open Journal of Instrumentation & Measurement

Advances In Materials and Processing Technologies

CONFERENCE REVIEWER

The 28th International Workshop on Electromagnetic Nondestructive Evaluation

The 22nd International Symposium on Applied Electromagnetics and Mechanics

PROFESSIONAL MEMBERSHIP

IEEE Student Member

ASNT Student Member
